

Project Demo Planning

Date	2 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Problem Statement & Flood Crisis Context	Present the real-world flood disaster statistics in India, annual economic losses, and the gap in existing early warning systems. Explain why ML-driven prediction is critical for proactive disaster management.	1	Sai Anjani Induru, Sameer Kumar, Sreekar Arveti, Nehanth Sanka, Sandeep Kore
2	System Architecture & Tech Stack Walkthrough	Walk through the three-tier architecture diagram — Flask backend, SQLite WAL database, and glassmorphism frontend. Highlight the XGBoost model selection rationale (483KB vs 234MB Random Forest) and StandardScaler pipeline.	1	Sai Anjani Induru, Sameer Kumar, Sreekar Arveti, Nehanth Sanka, Sandeep Kore
3	Live Flood Risk Prediction Demo	Perform a live end-to-end prediction: log in as a Meteorologist, input 10 weather parameters for Assam (high-risk region), submit the form, and show the real-time red flood warning alert card with probability percentage.	1	Sai Anjani Induru, Sameer Kumar, Sreekar Arveti, Nehanth Sanka, Sandeep Kore
4	RBAC Multi-Role Access Control Demo	Demonstrate the four user roles by logging in as Admin, Meteorologist, Local Authority, and Normal User sequentially. Show how each role has scoped permissions — Admin sees model analytics, Meteorologist accesses prediction forms, etc.	1	Sai Anjani Induru, Sameer Kumar, Sreekar Arveti, Nehanth Sanka, Sandeep Kore
5	Prediction Results & History	Show prediction results, risk level, prediction history, and explain how previous records are stored and retrieved.	1	Sai Anjani Induru, Sameer Kumar, Sreekar Arveti, Nehanth Sanka, Sandeep Kore
6	Future Scope & Conclusion	Explain future enhancements such as real-time weather API integration, GIS	1	Sai Anjani Induru, Sameer Kumar, Sreekar Arveti,

		mapping, SMS alerts, followed by project conclusion and Q&A.		Nehanath Sanka, Sandeep Kore
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Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Open with India's annual flood damage statistics (over \$3 billion losses). Explain the motivation for building a real-time ML-based early warning system. Briefly introduce the team and individual roles.
2	Solution Overview	Present the architecture diagram, tech stack (Flask + XGBoost + SQLite WAL + Glassmorphism CSS), and the ML pipeline flow from raw meteorological data to a real-time risk classification endpoint.
3	Live Feature Demonstration	Walk through the full user journey: Registration → Login → Enter 10 weather parameters for a high-risk state → View flood risk alert → Check prediction history → Admin model analytics dashboard. Show the Render deployment URL live.
4	Q&A Session	Open the floor for evaluator questions. Prepared answers for: Why XGBoost over Random Forest? How does WAL mode solve concurrency? What is the StandardScaler formula? How are roles enforced in Flask? What are future scalability plans?